

The Semantic Model

LESSON 03 · GROUND IN KNOWLEDGE — STUDIO: KNOWLEDGE GRAPH & SEMANTIC MODEL

What you'll learn

- Understand why vector RAG fails at structured data queries and how text-to-SQL bridges the gap.
- Connect a data source (e.g. CSV or SQL database) to a Semantic Data Model in Studio.
- Configure a Documentation Agent to auto-generate English schema descriptions of tables and columns.
- Inject custom business logic into column descriptions to guide text-to-SQL generation.
- Build an agent that uses Data Query to query database tables and return accurate answers.

Key concepts

Text-to-SQL enables exact data retrieval where vector search fails. If a question requires counting, summing, or filtering columns, similarity search returns wrong or hallucinated values. A semantic model lets the agent generate and run SQL queries against the database instead.

TAKEAWAY

Text-to-SQL enables exact data retrieval where vector search fails.

A Documentation Agent auto-generates schema documentation to guide the query builder. To write accurate SQL, the LLM needs plain English context of what each table and column represents. A Documentation Agent reads the schema and drafts these descriptions automatically.

TAKEAWAY

A Documentation Agent auto-generates schema documentation to guide the query builder.

Business logic is injected directly into column descriptions to steer the agent's SQL logic. By tuning descriptions—such as defining `total` as `line total in USD, quantity times unit price`—you give the agent the precise context it needs to choose and compute columns correctly.

TAKEAWAY

Business logic is injected directly into column descriptions to steer the agent's SQL logic.

The Data Query feature allows the agent to run and return SQL computations in seconds. Once attached to a semantic model, the agent translates human questions into SQL, runs it against the connected source, and presents the structured calculation results.

TAKEAWAY

The Data Query feature allows the agent to run and return SQL computations in seconds.

In Studio

Studio feature: **Build → Knowledge → Semantic Data Model**. Here is the configuration and testing flow:

1. Connect a data source under **Data Connectors** (e.g., upload a CSV file or connect Postgres, MySQL, or BigQuery).
2. Go to **Build → Knowledge**, click **New**, and select **Semantic Data Model**.
3. Enter a name, select the connected data source, and choose a **Documentation Agent** to scan the schema.
4. Open the created model to review the auto-generated table and column descriptions.

5. Edit and refine column descriptions to embed specific formulas or business rules (e.g., specifying currency or calculation rules).
6. Attach the semantic model to your agent under the **Data Query** feature in the agent settings.
7. Test in the Playground by asking complex aggregation queries (e.g., *"Which customer has spent the most in total?"*) and trace the query execution in the activity panel.

TRY THIS

- 1 Match these two queries to the correct retrieval type (Vector RAG vs. Data Query):

- *"Give me a summary of our company's refund rules for late shipments."* - *"Calculate the total number of refunds processed for late shipments last month."*

- 1 Create a test database schema with a **price** and **discount** column. Write a description for the total field that instructs the agent how to calculate net sales.